Question No-1.

If the received frame is 11010110111110 and the CRC generating polynomial is 10011 then determine if there is as error in the received message. IF there is no error in the received message the find the user data.

<u>Solution:</u>

Received frame: 11010110111110

Generating polynomial: 10011

$x^4.1 + x^3.0 + x^2.0 + x^1.1 + x^0.1$

$x^4 + x + 1$

Hence the order is 4 . So, massage after 4 zero bits are appended: 110101101111100000

```
             1100001010000
      10011) 110101101111100000
             10011
              10011
              10011
               00001
               00000
                00010
                00000
                 00101
                 00000
                  01011
                  00000
                   10111
                   10011
                    01001
                    00000
                     10011
                     10011
                      00000
                      00000
                       00000
                       00000
                        00000
                        00000
                         00000
                         00000
                          00000
                          00000
                           0000
```

Remainder is zero. So, there is no error.

User data is: 11010110111110

Question No-2.
A fifteen bit hamming code is received as 110101100011001. What is the correct code?

Solution:

The received hamming code is: 110101100011001
Finding parity bits P_1, P_2 , P_4 , P_8 of received code.

Check parity bit of: P_1 = ?0010101 = Yes
Check parity bit of: P_2 = ?0110101 = No
Check parity bit of: P_4 = ?0111001 = No
Check parity bit of: P_8 = ?0011001 = No

Error bit is: 2+4+8 = 14

The 14th position of the received bit is error.
The correct code is: 1101011000110011

Question No-3.
The received code word is 1100100101011, check if there is error in the codeword if the divisor is 10101.

Solution:
Dividing the code word by 10101, we get

```
              111110001
      10101) 1100100101011
             10101
              11000
              10101
               11010
               10101
                11111
                10101
                 10100
                 10101
                  11011
                  10101
                   1110
```

Remainder is 1110.
Non-zero remainder indicates that there are errors in the received code word.

Question No-4.
Compute the checksum in CRC and calculate the CRC for the frame 110101011 and generate Polynomial x4 +x+1 and write the transmitted frame also check the error with transmitted Frame.

Solution:
Here we have to convert Generator polynomial into equivalent bits pattern. [Generator polynomial is nothing but divisor]

Polynomial is X^4+x+1
= $X^4.1+X^3.0+X^2.0+X^1.1+X^0.1$
= 10011

The order is 4. So, the message after 4 zero are appended
1101010110000

```
              110000011
10011) 1101010110000
       10011
        10011
        10011
         00000
         00000
          00001
          00000
           00011
           00000
            00110
            00000
             01100
             00000
              11000
              10011
               10110
               10011
                01010
                00000
                 1010
```

The CRC code is 0101.
Non-zero reminder indicates that there are error in the given frame.
The correct frame is: 1101010110101

Question No-5.

Assuming even parity, find the parity bit for each of the following data units.

a. 1001011
b. 0001100
c. 1000000
d. 1110111

Solution:

Sl. No.	Data word	Number of 1s	Parity	Code Word
a	1001011	4 (Even)	0	0 1001011
b	0001100	2 (Even)	0	0 0001100
c	1000000	1(Odd)	1	1 1000000
d	1110111	6 (Even)	0	0 1110111

Question No-6.

Given the dataword 1010011110 and the divisor 10111,

a. Show the generation of the codeword at the sender site (using binary division).
b. Show the checking of the codeword at the receiver site (assume no error).

Solution:

The dataword is 1010011110 and the divisor 10111.
The message bit after 4 zero are appended 10100111100000

(a) Binary division case:

```
               1001101110
        10111) 10100111100000
               10111
                00111
                00000
                 01111
                 00000
                  11111
                  10111
                   10001
                   10111
                    01100
                    00000
                     11000
                     10111
                      11110
                      10111
                       10010
                       10111
                        01010
                        00000
                         1010
```

CRC Checksum was 1010
Codeword was 10100111101010

(b) Receiver using binary division:

```
10111)10100111101010
      10111
       00111
       00000
        01111
        00000
         11111
         10111
          10001
          10111
           01100
           00000
            11001
            10111
             11100
             10111
              10111
              10111
               00000
               00000
                0000
```

Reminder is 0000
So code has no error.

Question No-7.
Using the code in Table 10.2, what is the data word if one of the following code words is received?

a. 01011
b. 11111
c. 00000
d. 11011

Solution:

Sl. No.	Code Word	Data Word
a	01011	01
b	11111	1111
c	00000	00
d	11011	1101